

# The MHD rotor in JEXPRESSO: the exact algorithm

problems/MHD/rotorDaoNazarov2022 — discretization, DYNSGS setup, constants, and positivity

## Abstract

This note states, term by term, the algorithm that runs when the deck `problems/MHD/rotorDaoNazarov2022/user_inputs.jl` is executed: the GLM-MHD equations and initial data, the spectral-element discretization, the low-storage Runge–Kutta integrator, the dynamic sub-grid-scale (DYNSGS) artificial viscosity exactly as configured, the conservative positivity limiter that keeps density and pressure positive, and the meaning and value of every constant. File and line references are to the repository at the commit that adds this note. The last sections give the measured results and what the algorithm does not guarantee.

## Contents

|          |                                                  |           |
|----------|--------------------------------------------------|-----------|
| <b>1</b> | <b>The problem</b>                               | <b>1</b>  |
| <b>2</b> | <b>Spatial discretization</b>                    | <b>3</b>  |
| <b>3</b> | <b>Time integration</b>                          | <b>3</b>  |
| <b>4</b> | <b>DynSGS as configured</b>                      | <b>4</b>  |
| 4.1      | History and hold . . . . .                       | 4         |
| 4.2      | The legacy residual . . . . .                    | 4         |
| 4.3      | Element normalization . . . . .                  | 5         |
| 4.4      | Coefficient . . . . .                            | 5         |
| 4.5      | Viscous operator . . . . .                       | 5         |
| <b>5</b> | <b>How positivity is enforced</b>                | <b>6</b>  |
| <b>6</b> | <b>Constants and deck keys</b>                   | <b>8</b>  |
| <b>7</b> | <b>Results</b>                                   | <b>9</b>  |
| <b>8</b> | <b>Differences from Dao &amp; Nazarov (2022)</b> | <b>10</b> |

## 1 The problem

**State.** The nine conserved fields (“slots”) are

$$\mathbf{q} = (\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi), \quad \mathbf{m} = (\rho u, \rho v, \rho w), \quad \mathbf{v} = \mathbf{m}/\rho, \quad (1)$$

in this order (`initialize.jl`, `qvars`).  $E$  is the total energy, including the magnetic and GLM energies. The pressure is

$$p = (\gamma - 1) \left( E - \frac{|\mathbf{m}|^2}{2\rho} - \frac{1}{2}|\mathbf{B}|^2 - \frac{1}{2}\psi^2 \right), \quad \gamma = 1.4. \quad (2)$$

Magnetic pressure is  $\frac{1}{2}|\mathbf{B}|^2$  (Heaviside–Lorentz units, as in Tóth 2000).  $\rho w$  and  $B_z$  start at zero and stay zero in this 2D problem.

**Equations.** Ideal GLM-MHD with Dedner divergence cleaning, plus the DYN SGS regularization of Section 4:

$$\partial_t \mathbf{q} + \partial_x \mathbf{F}(\mathbf{q}) + \partial_y \mathbf{G}(\mathbf{q}) = \mathbf{S}(\mathbf{q}) + \nabla \cdot (\nu \nabla \mathbf{q}). \quad (3)$$

The fluxes are those of `problems/MHD/orszagTangBormanis2024/user_flux.jl` (included by the rotor’s `user_flux.jl`, which fixes  $\gamma = 1.4$ ). With  $p_{tot} = p + \frac{1}{2}|\mathbf{B}|^2$  and  $e_f = \frac{1}{2}\rho|\mathbf{v}|^2 + \gamma p/(\gamma - 1) + |\mathbf{B}|^2$ :

| slot     | $\mathbf{F}$ (x-flux)                                     | $\mathbf{G}$ (y-flux)                                     |
|----------|-----------------------------------------------------------|-----------------------------------------------------------|
| $\rho$   | $\rho u$                                                  | $\rho v$                                                  |
| $\rho u$ | $\rho u^2 + p_{tot} - B_x^2$                              | $\rho uv - B_x B_y$                                       |
| $\rho v$ | $\rho vu - B_x B_y$                                       | $\rho v^2 + p_{tot} - B_y^2$                              |
| $E$      | $u e_f - B_x(\mathbf{v} \cdot \mathbf{B}) + c_h \psi B_x$ | $v e_f - B_y(\mathbf{v} \cdot \mathbf{B}) + c_h \psi B_y$ |
| $\rho w$ | $\rho w u - B_x B_z$                                      | $\rho w v - B_y B_z$                                      |
| $B_x$    | $c_h \psi$                                                | $v B_x - u B_y$                                           |
| $B_y$    | $u B_y - v B_x$                                           | $c_h \psi$                                                |
| $B_z$    | $u B_z - w B_x$                                           | $v B_z - w B_y$                                           |
| $\psi$   | $c_h B_x$                                                 | $c_h B_y$                                                 |

There is no Powell source. The only source is the GLM damping  $S_\psi = -(c_h/c_r)\psi$  with  $c_r = 0.18$  (`user_source.jl`), a damping rate of  $c_h/c_r = 14.33$ .

**Cleaning speed.**  $c_h$  is computed once from the initial condition and held fixed (`initialize.jl`):

$$c_h = \max_{\text{nodes, all ranks}} \left( |\mathbf{v}| + \sqrt{(\gamma p + |\mathbf{B}|^2)/\rho} \right) = 2.5802306434. \quad (4)$$

**Domain, time, boundaries.**  $\Omega = (0, 1)^2$ , doubly periodic,  $t \in [0, 0.15]$ , output at  $t = 0, 0.05, 0.10, 0.15$ .

**Initial condition** (Balsara & Spicer 1999; Tóth 2000, first rotor; Dao & Nazarov 2022, §5.5). With  $r = |\mathbf{x} - (0.5, 0.5)|$ ,  $r_0 = 0.1$ ,  $r_1 = 0.115$ ,  $u_0 = 2$ ,  $f = (r_1 - r)/(r_1 - r_0)$ :

| region             | $\rho$   | angular velocity $\omega$ |
|--------------------|----------|---------------------------|
| $r < r_0$          | 10       | $u_0/r_0 = 20$            |
| $r_0 \leq r < r_1$ | $1 + 9f$ | $f u_0/r$                 |
| $r \geq r_1$       | 1        | 0                         |

$$(u, v) = \omega (0.5 - y, x - 0.5), \quad p = 1, \quad \mathbf{B} = (5/\sqrt{4\pi}, 0, 0), \quad \psi = 0, \quad E = p/(\gamma - 1) + \frac{1}{2}\rho|\mathbf{v}|^2 + \frac{1}{2}|\mathbf{B}|^2.$$

## 2 Spatial discretization

**Mesh.** The  $32 \times 32$  quadrilateral mesh `orszagTangBormanis2024/OT_32x32_periodic.msh` is refined uniformly by `p4est :init_refine_lvl = L` times:  $(32 \cdot 2^L)^2$  elements,  $128^2$  at  $L = 2$  (the deck) and  $256^2$  at  $L = 3$ . Periodicity is not a node merge: the two copies of a seam node keep their coordinates and local indices and share a global index (`ip2gip`); they are summed by the global assembly below.

**Basis and quadrature.** Lagrange polynomials of degree  $N = \text{:nop} = 3$  on the  $N + 1 = 4$  Legendre–Gauss–Lobatto (LGL) nodes per direction,  $385 \times 385$  nodes at  $L = 2$ . Quadrature uses the same LGL nodes (inexact integration), so the mass matrix is diagonal:

$$M_I = \sum_{K \ni I} m_I^K, \quad m_{ij}^K = \omega_i \omega_j J_{ij}^K, \quad (5)$$

summed over every element on every MPI rank and over both copies of periodic nodes (`element_matrices.jl`, `DSS_global_mass!`).

**Inviscid term** (strong form, collocated):

$$r_{ij}^K = -\omega_i \omega_j J_{ij}^K [\partial_x I_h \mathbf{F} + \partial_y I_h \mathbf{G} - \mathbf{S}]_{ij}, \quad (6)$$

where  $I_h$  is the degree- $N$  interpolant on the element and the derivatives are the LGL differentiation matrix applied to the nodal fluxes.

**Viscous term** (weak form, Section 4.5):

$$d_{ij}^K = -\sum_{k,l} \omega_k \omega_l J_{kl}^K \nabla \phi_{ij}(\mathbf{x}_{kl}) \cdot \nu_K \nabla \mathbf{q}(\mathbf{x}_{kl}). \quad (7)$$

There is no boundary integral (the domain is periodic).

**Assembly.** The element contributions are summed by local index (`DSS_rhs!`), then over ranks and periodic copies by `assemble_mpi!` (`DSS_global_RHS!`), and divided by the lumped mass:

$$\dot{\mathbf{q}}_I = M_I^{-1} \sum_{K \ni I} (r_I^K + d_I^K). \quad (8)$$

## 3 Time integration

**Scheme.** Carpenter–Kennedy five-stage, fourth-order, 2N low-storage Runge–Kutta (CarpenterKennedy2N54, OrdinaryDiffEqLowStorageRK). With  $k = \dot{\mathbf{q}}$  and registers  $(\mathbf{q}, \mathbf{t})$ , one step of size  $h$  is

$$\mathbf{t} \leftarrow h k(\mathbf{q}, t_n), \quad \mathbf{q} \leftarrow \mathbf{q} + B_1 \mathbf{t}, \quad (9)$$

$$\text{for } s = 2, \dots, 5: \quad \mathbf{t} \leftarrow A_s \mathbf{t}, \quad \boxed{\mathbf{q} \leftarrow \mathcal{L}(\mathbf{q})}, \quad \mathbf{t} \leftarrow \mathbf{t} + h k(\mathbf{q}, t_n + c_s h), \quad \mathbf{q} \leftarrow \mathbf{q} + B_s \mathbf{t}, \quad (10)$$

$$\boxed{\mathbf{q} \leftarrow \mathcal{L}(\mathbf{q})} \quad (\text{end of step}), \quad (11)$$

where  $\mathcal{L}$  is the positivity limiter of Section 5 (the “stage limiter”). The coefficients are

| $s$ | $A_s$                        | $B_s$                        | $c_s$                       |
|-----|------------------------------|------------------------------|-----------------------------|
| 1   | 0                            | 1432997174477/9575080441755  | 0                           |
| 2   | -567301805773/1357537059087  | 5161836677717/13612068292357 | 1432997174477/9575080441755 |
| 3   | -2404267990393/2016746695238 | 1720146321549/2090206949498  | 2526269341429/6820363962896 |
| 4   | -3550918686646/2091501179385 | 3134564353537/4481467310338  | 2006345519317/3224310063776 |
| 5   | -1275806237668/842570457699  | 2277821191437/14882151754819 | 2802321613138/2924317926251 |

i.e.  $c = (0, 0.149659, 0.370401, 0.622256, 0.958282)$ , with Butcher weights  $b = (0.005594, 0.344743, 0.028912, 0.467694, 0.153057)$ ,  $\sum_s b_s c_s = \frac{1}{2}$ .

**Step.**  $\Delta t = : \Delta t = 2 \times 10^{-4}$  at  $L = 2$  (acoustic CFL  $\approx 0.24$  against the smallest LGL spacing  $2.16 \times 10^{-3}$ ). The integrator is handed `Float32( $\Delta t$ ) = 1.99999995  $\times 10^{-4}$` , and 250 such steps end  $1.26 \times 10^{-9}$  short of each output time, which one extra step of that length then reaches; DYN SGS uses `params. $\Delta t$`  in double precision.

**Limiter hook.** `:lpositivity => true` with `:positivity_method => "conservative"` makes `mod_inputs.jl` rebuild the solver as `CarpenterKennedy2N54(stage_limiter! = positivity_conservative_limiter!, williamson_condition = false)`. The flag `williamson_condition = false` is required: with the default `true` the package fuses stages 2–5 and never calls the stage limiter there.  $\mathcal{L}$  therefore runs 5 times per step, the same number on every rank.

## 4 DynSGS as configured

The model is `:visc_model => DSGS_MHD()` with `:lvisc => true`, in its conserved form (`:dsgs_conserved => true`), with the legacy sensor (`:dsgs_sensor => "legacy"`) and element norms (`:dsgs_norms => "element"`). The coefficient is one number  $\nu_K$  per element, recomputed at every Runge–Kutta stage (`src/kernel/physics/SGS.jl, compute_dsgs_viscosity!`).

### 4.1 History and hold

DYN SGS keeps its own three-level history ( $\mathbf{q}^n, \mathbf{q}^{n-1}, \mathbf{q}^{n-2}$ ). At the first right-hand-side call of each step ( $t - t_{hist} \geq 0.999 \Delta t$ ) it rotates:  $\mathbf{q}^{n-2} \leftarrow \mathbf{q}^{n-1}$ ,  $\mathbf{q}^{n-1} \leftarrow \mathbf{q}^n$ ,  $\mathbf{q}^n \leftarrow \mathbf{q}$  (the current, already limited, state),  $t_{hist} \leftarrow t$  (`rhs.jl`, rotation block). With `:dsgs_hold_steps => 2` the coefficient is held at zero until three rotations have happened:  $\nu = 0$  during the first two steps, when the history is still the initial condition.

### 4.2 The legacy residual

At every stage of every later step, for node  $I$  (local index of node  $(i, j)$  of element  $K$ ) and slot  $q = 1, \dots, 8$  ( $\psi$  is excluded),

$$R_{K,ij}^q = \left| \frac{3q_I - 4q_I^n + q_I^{n-1}}{2\Delta t} - (M^{-1}\mathcal{R}_{inv})_I^q \right|, \quad (12)$$

where  $q_I$  is the current stage value and  $\mathcal{R}_{inv} = \sum_K r^K$  is the inviscid right-hand side, *assembled over all ranks and periodic copies*: a copy of the rank-local sum is passed through `assemble_mpi!` before it is used (`rhs.jl, _dsgs_residual_rhs!`). Without that step the spatial term at rank-interface and periodic-seam nodes would be only the fraction  $(\frac{1}{2}, \frac{1}{4})$  of the rate that the rank's own elements contribute, and  $\nu$  would depend on the MPI partition.

**What the legacy residual measures.** With  $\mathbf{q}_{stage} - \mathbf{q}^n = c_s \Delta t \dot{\mathbf{q}} + O(\Delta t^2)$  and  $\mathbf{q}^n - \mathbf{q}^{n-1} = \Delta t \dot{\mathbf{q}} + O(\Delta t^2)$ ,

$$R_s = |(1.5c_s - 1.5)F + (1.5c_s - 0.5)V| + O(\Delta t), \quad (13)$$

with  $F$  the inviscid and  $V$  the viscous rate. Where  $V \ll F$  this is  $1.5(1 - c_s)|\dot{\mathbf{q}}|$ ; weighted with the Butcher weights,  $\sum_s b_s 1.5(1 - c_s) = 0.75$ . The legacy residual is therefore a flux-divergence (gradient) sensor, not a truncation-error residual, and it does not depend on  $\Delta t$ : a smaller step converges to the same dissipated solution.

### 4.3 Element normalization

Over the 16 nodes of element  $K$  (plain node averages  $\langle \cdot \rangle_K$ ):

$$S_K^q = \max_{I \in K} |q_I - \langle q \rangle_K|, \quad \bar{\rho}_K = \max(|\langle \rho \rangle_K|, \varepsilon), \quad (14)$$

$$\bar{p}_K = (\gamma - 1) \max \left( \langle E \rangle_K - \frac{|\langle \mathbf{m} \rangle_K|^2}{2\bar{\rho}_K} - \frac{1}{2} |\langle \mathbf{B} \rangle_K|^2, 0 \right), \quad \bar{c}_K = \sqrt{\max(\gamma \bar{p}_K / \bar{\rho}_K, \varepsilon)}, \quad (15)$$

and the normalizer of slot  $q$  is

$$n_K^q = \max(S_K^q, r_{loc} s_K^q) + \varepsilon, \quad s^\rho = \bar{\rho}, \quad s^{\rho u, \rho v, \rho w} = \bar{\rho} \bar{c}, \quad s^E = \bar{\rho} \bar{c}^2, \quad s^B = \sqrt{\bar{\rho}} \bar{c}, \quad (16)$$

with  $r_{loc} = \text{:dsgs\_local\_rel} = 1$  and  $\varepsilon = 10^{-16}$ . Each element is thus judged against its own spread and its own physical scales, not the domain's: in the low-pressure interior of the rotor ( $p \ll \frac{1}{2}\rho|\mathbf{v}|^2$ ) domain-wide norms made the residual of grid-scale pressure noise invisible.

### 4.4 Coefficient

$$\mathcal{R}_K = \max_{I \in K} \max_{q \leq 8} \frac{R_{K,I}^q}{n_K^q}, \quad w_K = \max_{I \in K} \left( |\mathbf{v}_I| + \sqrt{\max(\gamma p_I / \rho_I + |\mathbf{B}_I|^2 / \rho_I, 0)} \right), \quad (17)$$

$$\Delta_K = \frac{\Delta_K^{elem}}{N + 1}, \quad \nu_K = \max \left( 0, \min(C_{\max} \Delta_K w_K, C_R \Delta_K^2 \mathcal{R}_K) \right), \quad (18)$$

with  $\Delta_K^{elem}$  the shortest edge of  $K$  (1/128, so  $\Delta_K = 1/512$ ),  $C_R = \text{:dsgs\_CR} = 1$ ,  $C_{\max} = \text{:dsgs\_Cmax} = 0.5$ , no lower floor ( $\text{:dsgs\_Cmin} = 0$ ) and no cutoff. In  $w_K$ ,  $\rho_I$  is floored at  $\varepsilon$  and  $p_I$  is the full nodal pressure (2). The cap  $C_{\max} \Delta_K w_K$  is the first-order (Lax–Friedrichs-like) viscosity; the residual term is  $O(\Delta^2)$  in smooth flow. With  $:\mu = (1, \dots, 1)$  every slot gets  $\nu_K$ .

### 4.5 Viscous operator

In the conserved form the diffused variables are the conserved variables themselves (`user_primitives.jl` copies  $\mathbf{q}$ ), and every slot, momentum included, gets the same flux

$$\mathbf{F}_V(q) = \nu_K \nabla q, \quad q \in \{\rho, \rho u, \rho v, E, \rho w, B_x, B_y, B_z, \psi\}, \quad (19)$$

assembled in the weak form of Section 2 (`rhs.jl`, `_expansion_visc!`, `flag lcons`). There is no  $\boldsymbol{\tau} \cdot \mathbf{v}$  work term and no separate heat flux: the regularization is the monolithic parabolic one,  $\partial_t \mathbf{q} + \dots = \nabla \cdot (\nu \nabla \mathbf{q})$ .

**Why momentum must get a Laplacian.** For any  $\nu \geq 0$ , the chain rule and the convexity of  $|\mathbf{m}|^2/(2\rho)$  and  $\frac{1}{2}|\mathbf{B}|^2$  give, for the regularization (19) alone,

$$\partial_t(\rho e) = \nabla \cdot (\nu \nabla(\rho e)) + \nu(\rho |\nabla \mathbf{v}|^2 + |\nabla \mathbf{B}|^2 + |\nabla \psi|^2), \quad \rho e = E - \frac{|\mathbf{m}|^2}{2\rho} - \frac{1}{2}|\mathbf{B}|^2 - \frac{1}{2}\psi^2, \quad (20)$$

and  $\partial_t \rho = \nabla \cdot (\nu \nabla \rho)$ : density and internal energy obey a minimum principle. Before this version the momentum slots received the deviatoric stress  $\nu(\nabla \mathbf{m} + \nabla \mathbf{m}^T - \frac{2}{3}(\nabla \cdot \mathbf{m})I)$ , which for constant  $\nu$  equals  $\nu[\Delta \mathbf{m} + \frac{1}{3}\nabla(\nabla \cdot \mathbf{m})]$  and adds the sign-indefinite term  $-\frac{\nu}{3}\mathbf{v} \cdot \nabla(\nabla \cdot \mathbf{m})$  to the equation above. In the rotor,  $\frac{1}{2}\rho|\mathbf{v}|^2$  reaches 20 where  $\rho e \approx 0.1$ . The minimum principle is a property of the PDE; the high-order discrete operator does not inherit it, which is why the limiter of Section 5 exists.

## 5 How positivity is enforced

Positivity of  $\rho$  and  $p$  is enforced by the conservative element limiter  $\mathcal{L}$  (`src/kernel/positivity/Positivity.jl`, functions `cpos_*`; driver `positivity_conservative_limiter!` in `positivity_driver.jl`), applied to the Runge–Kutta state before stages 2–5 and at the end of every step (Section 3). It acts on the hydrodynamic part  $(\rho, \mathbf{m}, \tilde{E})$  with

$$\tilde{E} = E - \frac{1}{2}|\mathbf{B}|^2 - \frac{1}{2}\psi^2, \quad P(\rho, \mathbf{m}, \tilde{E}) = (\gamma - 1)\left(\tilde{E} - \frac{|\mathbf{m}|^2}{2\rho}\right) = p, \quad (21)$$

and never changes  $\mathbf{B}$  or  $\psi$ .  $P$  is concave in  $(\rho, \mathbf{m}, \tilde{E})$  for  $\rho > 0$  ( $|\mathbf{m}|^2/(2\rho)$  is the perspective of a convex function). The admissible set is  $G = \{\rho \geq \varepsilon_\rho, P \geq \varepsilon_p\}$ , which is convex.

**Constants.**  $\varepsilon_\rho = \text{:positivity\_rho\_min} = 10^{-6}$ ,  $\varepsilon_p = \text{:positivity\_p\_min} = 10^{-6}$  (absolute;  $10^{-6}$  of the ambient  $\rho = p = 1$ ),  $\gamma = \text{:dsgs\_gamma} = 1.4$ , and a round-off margin factor  $s = 64$  (ulps).

**Step 0: gate.** Every rank counts its nodes outside  $G$  (NaN counts as outside) and the counts are summed with one `MPI_Allreduce`. If the sum is zero,  $\mathcal{L}$  returns and the state is bitwise unchanged.

**Step 1: element means and admissibility.** For every element  $K$  holding a node outside  $G$ , with the lumped weights  $m_i = \omega_a \omega_b J_{ab}^K$  of (5) and  $W = \sum_i m_i$ ,

$$\bar{\rho} = \frac{1}{W} \sum_i m_i \rho_i, \quad \bar{\mathbf{m}} = \frac{1}{W} \sum_i m_i \mathbf{m}_i, \quad \bar{E} = \frac{1}{W} \sum_i m_i \tilde{E}_i. \quad (22)$$

With  $R = \max(\max_i |\rho_i|, |\bar{\rho}|)$ ,  $\hat{E} = \max_i (|E_i| + \frac{1}{2}|\mathbf{B}_i|^2 + \frac{1}{2}\psi_i^2)$ ,

$$\delta = s \text{ulp}(R), \quad \bar{\rho}_g = \bar{\rho} - \delta, \quad \varepsilon_\rho^+ = \varepsilon_\rho + \delta, \quad \varepsilon_p^+ = \varepsilon_p + s(\gamma - 1) \text{ulp}(\hat{E}), \quad \bar{p} = P(\bar{\rho}_g, \bar{\mathbf{m}}, \bar{E}). \quad (23)$$

The element is limited only if  $\bar{\rho}_g > \varepsilon_\rho^+$  and  $\bar{p} > \varepsilon_p^+$  (both finite); otherwise it is counted as failed and left to step 5.

**Step 2: density.** If  $\rho_{\min} = \min_i \rho_i < \varepsilon_\rho^+$ ,

$$\theta_\rho = \frac{\bar{\rho} - \varepsilon_\rho^+}{\bar{\rho} - \rho_{\min}}, \quad \tilde{\rho}_i = \bar{\rho} + \theta_\rho(\rho_i - \bar{\rho}); \quad (24)$$

otherwise  $\theta_\rho = 1$ ,  $\tilde{\rho}_i = \rho_i$ . Then  $\min_i \tilde{\rho}_i \geq \varepsilon_\rho^+$ .

**Step 3: pressure (root rule).** For each node with  $P(\tilde{\rho}_i^g, \mathbf{m}_i, \tilde{E}_i) < \varepsilon_p^+$ , where  $\tilde{\rho}_i^g = \tilde{\rho}_i - s \text{ulp}(\max(|\rho_i|, \bar{\rho}))$ , let  $t_i$  be the first  $t \in (0, 1]$  at which the segment from  $(\bar{\rho}_g, \bar{\mathbf{m}}, \bar{E})$  to  $(\tilde{\rho}_i^g, \mathbf{m}_i, \tilde{E}_i)$  reaches  $P = \varepsilon_p^+$ . Writing  $e^+ = \varepsilon_p^+ / (\gamma - 1)$ ,  $\Delta\rho = \tilde{\rho}_i^g - \bar{\rho}_g$ ,  $\Delta\mathbf{m} = \mathbf{m}_i - \bar{\mathbf{m}}$ ,  $\Delta E = \tilde{E}_i - \bar{E}$ ,  $t_i$  is the root in  $(0, 1)$  of  $at^2 + bt + c = 0$  with

$$a = \Delta\rho \Delta E - \frac{1}{2}|\Delta\mathbf{m}|^2, \quad b = \bar{\rho}_g \Delta E + \Delta\rho(\bar{E} - e^+) - \bar{\mathbf{m}} \cdot \Delta\mathbf{m}, \quad c = \bar{\rho}_g(\bar{E} - e^+) - \frac{1}{2}|\bar{\mathbf{m}}|^2 > 0, \quad (25)$$

$$t_i = \begin{cases} 2c/(\sqrt{b^2 - 4ac} - b), & b \leq 0, \\ (-b - \sqrt{b^2 - 4ac})/(2a), & b > 0, a < 0, \\ 1, & b > 0, a \geq 0, \end{cases} \quad (26)$$

clipped to  $[0, 1]$  (a negative or NaN value gives 0;  $\tilde{\rho}_i^g \leq 0$  gives 0). All other nodes have  $t_i = 1$ , and  $\theta_p = \min_i t_i$ . (Concavity guarantees that the chord value  $(\bar{p} - \varepsilon_p^+)/(\bar{p} - P_i)$  is a lower bound of  $t_i$ ; the root is used because the chord flattens whole elements where density limiting binds.)

**Step 4: conservative update.** The limited element copy is  $\mathbf{q}_i^K = \bar{\mathbf{q}} + \theta_p(\tilde{\mathbf{q}}_i - \bar{\mathbf{q}})$  in  $(\rho, \mathbf{m}, \tilde{E})$ ; its increments

$$\delta\rho_i = (\theta_p \theta_\rho - 1)(\rho_i - \bar{\rho}), \quad \delta\mathbf{m}_i = (\theta_p - 1)(\mathbf{m}_i - \bar{\mathbf{m}}), \quad \delta E_i = (\theta_p - 1)(\tilde{E}_i - \bar{E}), \quad \delta\mathbf{B}_i = \delta\psi_i = 0 \quad (27)$$

are weighted,  $A_I += m_i \delta_i$ , assembled over ranks and periodic copies with `assemble_mpi!` (slots  $\rho, \rho u, \rho v, E, \rho w$ ), and applied as  $\mathbf{q}_I \leftarrow \mathbf{q}_I + M_I^{-1} A_I$ . Equivalently,  $\mathbf{q}_I = \sum_{K \ni I} (m_I^K / M_I) \mathbf{q}_I^K$  with  $\mathbf{q}_I^K = \mathbf{q}_I$  for elements that were not limited.

**Step 5: fallback.** Nodes still outside  $G$  can only come from failed elements (inadmissible mean). On every rank such a node gets the floor:  $\rho \leftarrow \varepsilon_\rho + s \text{ulp}(\varepsilon_\rho)$  if  $\rho < \varepsilon_\rho$ ; then, if  $\tilde{E} - k < e_{\min}$ , the momentum is scaled by  $\theta = \sqrt{(\tilde{E} - e_{\min})/k}$ ,  $k = |\mathbf{m}|^2/(2\rho)$ ,  $e_{\min} = \varepsilon_p/(\gamma - 1) + s \text{ulp}(|E| + \frac{1}{2}|\mathbf{B}|^2 + \frac{1}{2}\psi^2)$  (total energy unchanged), or, if  $\tilde{E} \leq e_{\min}$ , set to zero with  $E$  raised so that  $\tilde{E} = e_{\min}$ . This step does *not* conserve mass, momentum or energy; every use is counted and reported.

### What it guarantees.

- Conservation:  $\sum_I M_I q_I$  over unique nodes is unchanged by steps 1–4 for all nine slots, because  $\sum_i m_i(x_i - \bar{x}) = 0$  in every element.
- Bounds: if every flagged element passes step 1, every node ends in  $G$  (each copy is in  $G$ ,  $G$  is convex, the reassembly is a convex combination).
- $\mathbf{B}$ ,  $\psi$  and the discrete  $\nabla \cdot \mathbf{B}$  are untouched;  $E$  changes only by exchanging kinetic and internal energy inside the element.
- Locality: only nodes of limited elements (and floored nodes) change; the result is the same on any number of ranks (checked bitwise on 1 and 3 ranks).

**What it does not guarantee.** Continuous Galerkin with a lumped mass has no theorem that keeps element means admissible, with any Runge–Kutta method; step 5 exists for that case. The limiter enforces  $p \geq \varepsilon_p$ , not a physical pressure: a node pulled up to  $p = 10^{-6}$  still shows a large Mach number. It is not invariant-domain preserving and not entropy-consistent.

**Reporting.** Every `:positivity_report_every` limiter calls (default 1000) the ranks reduce the counters and print the number of limited elements, the smallest  $\theta_\rho$  and  $\theta_p$ , the failed elements and the floored nodes, if any of them grew.

## Verification.

- The Julia kernel reproduces an independent Python implementation of steps 0–4 to  $\leq 7 \times 10^{-14}$  on nine states: a  $128^2$  rotor state at  $t = 0.15$  from an earlier run, with  $\varepsilon_p = 0.03$  (94 elements limited), synthetic negative-density and negative-pressure states, a periodic-seam state, a near-vacuum element mean (sent to step 5), round-off stress states, and the  $256^2$  state at  $t = 0.1$  where the unlimited scheme had  $p = -0.0026$  at four nodes (4 elements limited,  $\theta_p \geq 0.9958$ ). In every case:  $\min \rho \geq \varepsilon_\rho$ ,  $\min p \geq \varepsilon_p$ ,  $|\Delta \sum_I M_I q_I| \leq 2 \times 10^{-16} \sum_I M_I |q_I|$  over steps 1–4,  $\mathbf{B}$  and  $\psi$  unchanged, periodic copies identical, and a second call changes nothing.
- On 3 MPI ranks, with rank boundaries and the periodic seam running through limited elements, the result is bitwise equal to the serial one.
- In a full run ( $128^2$ , with  $\varepsilon_p$  raised to 0.1 to force activity) the limiter made more than 150 element limitings between  $t = 0.05$  and 0.1 ( $\theta_p \geq 0.9978$ , no failed element, no floor), the minimum pressure at  $t = 0.1$  was 0.1, and the discrete mass and energy changed by  $3 \times 10^{-16}$  and  $1 \times 10^{-16}$  (relative).

## 6 Constants and deck keys

| key / symbol                                               | value              | meaning                                                               |
|------------------------------------------------------------|--------------------|-----------------------------------------------------------------------|
| <code>:ode_solver</code>                                   | CK(5,4) 2N         | low-storage RK, Section 3; rebuilt with the limiter                   |
| <code>:Δt</code>                                           | $2 \times 10^{-4}$ | time step ( $10^{-4}$ at $L = 3$ , $N = 3$ )                          |
| <code>:tinit</code> , <code>:tend</code>                   | 0, 0.15            | time interval                                                         |
| <code>:diagnostics_at_times</code>                         | 0, .05, .1, .15    | output times (also integrator stops)                                  |
| <code>:lsource</code>                                      | true               | GLM damping $S_\psi = -(c_h/c_r)\psi$                                 |
| $c_r$                                                      | 0.18               | Dedner damping ratio ( <code>glm_cr_mhd</code> )                      |
| $c_h$                                                      | 2.5802306434       | cleaning speed, from the initial data, constant                       |
| $\gamma$ ( <code>γ_mhd</code> , <code>:dsgs_gamma</code> ) | 1.4                | ratio of specific heats; the two must agree                           |
| <code>:SOL_VARS_TYPE</code>                                | TOTAL()            | the state is $\mathbf{q}$ itself, not a perturbation                  |
| <code>:nop</code>                                          | 3                  | polynomial degree $N$ ; $N + 1 = 4$ LGL nodes per direction           |
| <code>:init_refine_lvl</code>                              | 2                  | p4est refinements of the $32^2$ mesh: $128^2$ elements                |
| <code>:lvisc</code>                                        | true               | DYNSGS viscous term on                                                |
| <code>:visc_model</code>                                   | DSGS_MHD()         | DYNSGS for the 9-slot MHD state                                       |
| $\mu$                                                      | (1, ..., 1)        | per-slot multiplier of $\nu_K$                                        |
| <code>:dsgs_sensor</code>                                  | "legacy"           | residual (12), BDF2 on $(\mathbf{q}, \mathbf{q}^n, \mathbf{q}^{n-1})$ |
| <code>:dsgs_hold_steps</code>                              | 2                  | $\nu = 0$ in the first 2 steps                                        |
| <code>:dsgs_norms</code>                                   | "element"          | per-element normalization                                             |
| <code>:dsgs_local_rel</code> ( $r_{loc}$ )                 | 1 (default)        | weight of the physical floors $s_K^q$                                 |
| <code>:dsgs_CR</code> ( $C_R$ )                            | 1                  | residual coefficient in (18)                                          |
| <code>:dsgs_Cmax</code> ( $C_{\max}$ )                     | 0.5                | first-order cap in (18)                                               |

| key / symbol                                  | value              | meaning                                                                  |
|-----------------------------------------------|--------------------|--------------------------------------------------------------------------|
| :dsgs_Cmin                                    | 0 (default)        | no background viscosity                                                  |
| :dsgs_conserved                               | true               | diffuse the conserved variables, $\mathbf{F}_V = \nu \nabla \mathbf{q}$  |
| :dsgs_rel, :dsgs_Prt                          | 1, 1               | inert in this configuration                                              |
| :dsgs_nazarov_energy                          | false              | no split energy flux                                                     |
| $\Delta_K$                                    | 1/512              | DYN SGS length, shortest edge/( $N + 1$ )                                |
| $\varepsilon$                                 | $10^{-16}$         | DYN SGS division guard                                                   |
| :lpositivity                                  | true               | positivity enforcement on                                                |
| :positivity_method                            | "conservative"     | the element limiter of Section 5                                         |
| :positivity_rho_min<br>( $\varepsilon_\rho$ ) | $10^{-6}$          | density bound                                                            |
| :positivity_p_min ( $\varepsilon_p$ )         | $10^{-6}$          | pressure bound                                                           |
| $s$                                           | 64                 | round-off margin (ulps) of the limiter                                   |
| :positivity_report_<br>every                  | 1000               | limiter calls between reports                                            |
| :conservation_every                           | 1                  | steps between samples of the totals<br>( <code>conservation.dat</code> ) |
| :conservation_slots                           | [1, 4]             | the totals sampled: mass and total energy                                |
| :energy_equation                              | "energy"           | slot 4 is total energy                                                   |
| :lkep, :entropy_<br>variables                 | false              | pointwise (non-split) fluxes                                             |
| :linitial_refine,<br>:ladapt, :lfilter        | true, false, false | uniform refinement only; no adaptivity, no filter                        |

## 7 Results

All runs:  $N = 3, 4$  MPI ranks, the deck of this note (Laplacian on  $\rho \mathbf{v}$ , assembled legacy sensor, element norms,  $C_R = 1$ ,  $C_{\max} = 0.5$ , limiter with  $\varepsilon_\rho = \varepsilon_p = 10^{-6}$ ). "Noise" is the median (95th percentile) of the node-to-node second difference of  $p$  relative to  $p$  for  $r < 0.08$ ; "symmetry" is the largest relative deviation from the  $180^\circ$  point symmetry of the problem; conservation is the change of the discrete totals  $\sum_I M_{IqI}$ .

| grid, $\Delta t$          | $t$  | $\rho$        | $p$            | $\frac{1}{2} \mathbf{B} ^2$ | max Mach | limiter  |
|---------------------------|------|---------------|----------------|-----------------------------|----------|----------|
| $128^2, 2 \times 10^{-4}$ | 0.05 | 0.848 – 14.11 | 0.384 – 2.505  | 0.123 – 4.161               | 4.28     | idle     |
|                           | 0.10 | 0.730 – 12.62 | 0.076 – 2.110  | 0.081 – 3.402               | 4.39     | idle     |
|                           | 0.15 | 0.565 – 10.65 | 0.0295 – 1.969 | 0.0746 – 2.543              | 3.55     | idle     |
| $256^2, 10^{-4}$          | 0.05 | 0.833 – 15.89 | 0.336 – 2.564  | 0.075 – 4.360               | 4.60     | idle     |
|                           | 0.10 | 0.662 – 13.54 | 0.0144 – 2.105 | 0.052 – 3.507               | 9.93     | idle     |
|                           | 0.15 | 0.502 – 12.25 | 0.0281 – 2.041 | 0.0505 – 2.601              | 3.90     | 1126 el. |

At  $128^2$  the limiter never acts (no node falls below  $10^{-6}$ ); the noise is 0.05% (0.19%) at  $t = 0.15$  and the symmetry error  $2 \times 10^{-10}$ . At  $256^2$  the minimum pressure at  $t = 0.1$  is 0.0144, at four interior nodes of four elements on the edge of the disc ( $r \approx 0.108$ , where  $\rho$  falls from 10 to 1 inside one element); there the Mach number is 9.9. Between  $t \approx 0.11$  and 0.13 the pressure at such nodes reached  $\varepsilon_p$  and the limiter made 1126 element limitings, all mild ( $\theta_p \geq 0.993$ ,  $\theta_\rho = 1$ ), with no element whose mean was inadmissible and no use of the floor. At  $t = 0.15$  the noise is 0.008% (0.04%) and the symmetry error  $2 \times 10^{-9}$ .

**Conservation in time.** With `:conservation_every => 1` the totals  $\mathcal{M}(t) = \sum_K \sum_{ij} m_{ij}^K \rho_{ij}$  and  $\mathcal{E}(t) = \sum_K \sum_{ij} m_{ij}^K E_{ij} (= \sum_I M_I q_I)$  are summed on every rank with compensated (Neumaier) summation, reduced over the ranks, and written at every step to `conservation.dat` (`tools/plot_conservation.py`). Figure 1: over the whole run  $|\Delta\mathcal{M}|/\mathcal{M} \leq 6.7 \times 10^{-16}$  ( $128^2$ ) and  $1.7 \times 10^{-15}$  ( $256^2$ ), and  $|\Delta\mathcal{E}|/\mathcal{E} \leq 1.2 \times 10^{-16}$  and  $2.3 \times 10^{-16}$ : both totals move by a few units in the last place, through the DYN SGS operator, the GLM source and every limiter step alike.

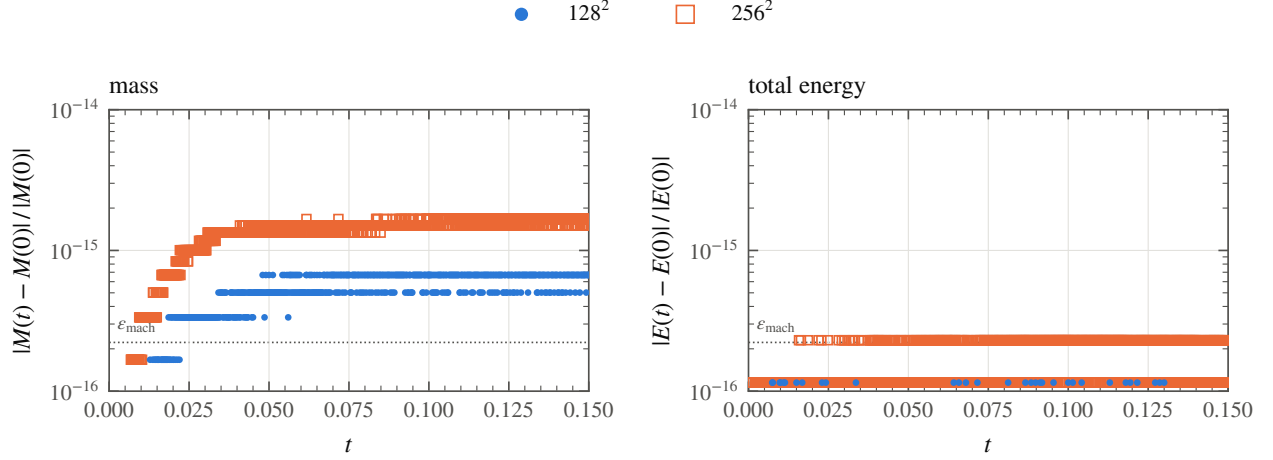


Figure 1: Relative change of the discrete mass and total energy at every time step,  $128^2$  (filled) and  $256^2$  (open) elements,  $N = 3$ . Samples with no change at all (exact zeros) cannot be drawn on the logarithmic axis and are omitted; the totals move by single units in the last place, hence the discrete levels. Dotted:  $\varepsilon_{\text{mach}}$ .

**Previous operator.** With the deviatoric stress on  $\rho \mathbf{v}$ , the rank-local legacy sensor and no limiter (commit `a3baa37`) the same deck gave, at  $128^2$ ,  $\min p = 0.059$  at  $t = 0.1$  and  $0.035$  at  $t = 0.15$  (max Mach 3.52), and at  $256^2$ ,  $p = -0.0026$  at four interior nodes at the edge of the disc at  $t = 0.1$  (Mach  $= \infty$  there), recovering to  $\min p = 0.032$  at  $t = 0.15$ .

**References at  $t = 0.15$ .** Tóth (2000, Fig. 18, flux-CT,  $400^2$ ):  $\rho \in [0.483, 12.95]$ ,  $p \in [0.0202, 2.008]$ ,  $\frac{1}{2}|\mathbf{B}|^2 \in [0.0177, 2.642]$ , max Mach 8.18 (at pressure undershoots). Dao & Nazarov (2022, Fig. 7, P3,  $300^2$  nodes):  $\rho \in [0.727, 8.42]$ ,  $p \in [0.0386, 1.93]$ ,  $\frac{1}{2}|\mathbf{B}|^2 \in [0.0551, 2.30]$ , max Mach 4.82.

## 8 Differences from Dao & Nazarov (2022)

- Regularization: DN22 use  $\nu \nabla \rho$  for mass and a Navier–Stokes/resistive flux for momentum, energy and  $\mathbf{B}$  (their eq. 4.4); here every conserved variable gets  $\nu \nabla q$ .
- Residual: DN22 evaluate  $|\text{BDF2}(\mathbf{q}^n, \mathbf{q}^{n-1}, \mathbf{q}^{n-2}) + \nabla \cdot \mathbf{F}(\mathbf{q}^n)|$  once per step, nodally, normalized by domain-wide variances; here the legacy sensor (12) is evaluated at every stage with element norms, and  $\nu$  is one value per element.
- Positivity: DN22 leave it outside the scope of their paper; here it is enforced by the conservative limiter of Section 5.

- Divergence: both use GLM cleaning.

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